Field Deployment Kit — Robust UAV Harvest-Readiness Monitoring

From bench to farm: assembling an industrial test run of the RobustIDPS.ai / M1+M4+M6+M7 network-layer robustness models on a crop-monitoring UAV.

- Mission: fly an agricultural field, capture multispectral imagery, decide **harvest-due / not-due per zone** — while the research models keep the UAV network layer (C2 link, telemetry, GNSS) alive under interference, jamming and spoofing.
- Value proposition proven on the bench: the M1+M4+M6+M7 stack sustains ≥ 90 % mission completion up to **~25 dB J/S** vs **~7.7 dB** undefended — so in a noisy/contested rural RF environment the drone still finishes the survey and still delivers the NDVI maps.

> Prices below are **indicative USD ranges** (2026), for planning only. Exact parts depend on field size, crop, region, and drone-law regime. Anything that transmits interference (jammer/spoofer for the EW test) is **strictly regulated** — see §11.

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1. System architecture — three tiers

The dissertation's three-tier resource budget maps directly onto hardware:

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TIER 1 — ONBOARD (on the UAV, flight-critical, lowest latency)
  Flight controller (PX4) ── companion edge AI module
    • M4 MambaShield (INT8)  → real-time C2/telemetry intrusion detection
    • fast NDVI/NDRE screen  → quick harvest-readiness flag per frame
    • anti-spoof GNSS + secure C2 radio

TIER 2 — EDGE GATEWAY (field base: van/tractor/tent, at the field edge)
  Rugged edge server / AGX-class box
    • Full RobustIDPS.ai stack
    • M1 CT/SDE-TGNN (temporal graph over telemetry/fleet)
    • M6 uncertainty calibration (trust score on detections)
    • M7 game-theoretic certification (decision policy + certificate)
    • Full crop-maturity model (multispectral → harvest-due map)
    • Federated-learning client, RTK base / NTRIP caster

TIER 3 — CLOUD / REGIONAL (office or cloud, offline-tolerant)
  GPU server
    • FedGTD federated aggregation across farms/fleets
    • PQC-signed model-update channel (Kyber-1024 / Dilithium)
    • Long-term storage, agronomy dashboards
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Which model runs where:

Model	Role	Tier	Why there
M4 MambaShield (INT8)	link/telemetry intrusion detection	1 onboard	lowest latency, flight-critical, cheap in TOPS
M1 CT/SDE-TGNN	temporal-graph anomaly over telemetry	2 edge	needs more compute, not per-packet-critical

Model	Role	Tier	Why there
M6 uncertainty calib.	trust score on detections	2 edge	feeds the decision, tolerates ~100 ms
M7 game-theoretic cert.	certified decision policy	2 edge / 3 cloud	offline-computable certificate
FedGTD	federated model aggregation	3 cloud	cross-farm, periodic
Harvest-readiness NDVI	quick per-zone maturity flag	1 onboard	immediate in-flight feedback
Crop-maturity model	full harvest-due map	2 edge	fuses calibrated reflectance + agronomy model

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2. The embedded chip (what you asked about specifically)

Primary onboard compute — recommended

NVIDIA Jetson Orin NX 16 GB (module) on a UAV-grade carrier board.

Spec	Value
AI throughput	~100 TOPS (INT8)
Power	10–25 W (configurable)
Weight	module + carrier ~45–120 g
Runs	MambaShield INT8 + NDVI inference **concurrently, real-time**
Software	JetPack/L4T, **TensorRT INT8** engines, Docker, MAVLink/ROS 2
Carriers	Seeed reComputer, ConnectTech Boson/Photon, Auvidea

Rationale: the M4 MambaShield model is quantised to **INT8** in the dissertation precisely so it fits an onboard budget like this; Orin NX has the headroom to also run the onboard NDVI screen and the video pipeline, at a power and weight a multirotor can carry.

Ultra-low-power alternative (security detector only)

If you want the intrusion detector on a **separate, ~2–3 W** co-processor and keep a lighter main CPU:

Option	TOPS	Power	Note
Hailo-8 M.2	26	~2.5 W	excellent perf/W for the INT8 detector
Google Coral Edge TPU	4	~2 W	cheapest, needs INT8 TFLite
Jetson **Orin Nano 8 GB**	~40	7–15 W	if you want one small NVIDIA module

Edge gateway (Tier 2) compute

Option	TOPS	Note
Jetson AGX Orin 64 GB	~275	single-box, rugged, runs full RobustIDPS.ai + M1/M6/M7
Industrial x86 + **RTX A2000/4000**	—	if you prefer x86 + more RAM/storage at the base

Cloud/regional (Tier 3)

Any GPU server (1× RTX 6000 Ada / L40S class) for FedGTD aggregation and PQC signing; can be a cloud VM.

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3. UAV platform & flight hardware

Item	Example	Indicative $
Airframe (multirotor, ≥ 30 min, payload ≥ 1.5 kg)	DJI M350 RTK, or Freefly Astro, or custom hexacopter	5,000–14,000
Flight controller (open, PX4/ArduPilot)	Holybro Pixhawk 6X / Cube Orange+	250–450
Companion computer	**Jetson Orin NX 16 GB + carrier** (see §2)	700–1,100
Redundant IMU / airspeed / barometer	(often integrated in FC)	—
Gimbal for payload	integrated or Gremsy	300–2,500
Spare props, arms, landing gear	—	200–500

> Fixed-wing/VTOL (e.g., WingtraOne, Quantum Trinity) is better for very large fields (>200 ha) — swap the airframe row, keep everything else.

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4. GNSS anti-spoof / anti-jam (the EW-critical part)

This is where the J/S results become real — pick a receiver with spoofing/jam detection, ideally the exact models the benchmark simulated:

Item	Example	Note
Multi-band RTK receiver	**u-blox ZED-F9P** (= benchmark "u-blox F9P")	cm-level RTK, spoof flags
— higher-assurance option	**Septentrio mosaic-X5** (AIM+ anti-jam/spoof)	strong EW resilience
— or	**NovAtel OEM7** (= benchmark "NovAtel OEM7")	matches benchmark model
Anti-jam antenna (serious EW)	CRPA / controlled-radiation-pattern antenna	nulls jammers
Downstream	receiver C/N_0 + spoof flags feed **M4/M6** as features	closes the loop

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5. Data links — the "network layer" the models monitor

Link	Example	Purpose
Primary C2 (encrypted MANET/mesh)	**Silvus StreamCaster**, Doodle Labs Smart Radio, or Microhard pDDL	the link M1/M4 inspect for intrusion
BVLOS backhaul	4G/5G modem (Sierra/Quectel) + NTRIP for RTK corrections	telemetry + updates
Telemetry radio (backup)	RFD900x	fail-safe C2
Ground antennas / mast / tracker	directional + omni, 3–6 m mast	range & link margin

The traffic across these radios is exactly what RobustIDPS.ai classifies — this is the "UAV networks layer" from the research.

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6. Agricultural sensing payload (harvest-due decision)

Item	Example	Gives you
Multispectral camera	**MicaSense Altum-PT** or RedEdge-P, Sentera 6X	NDVI, NDRE, red-edge → maturity

Item	Example	Gives you
Downwelling light sensor + calibrated panel	MicaSense DLS 2 + reflectance panel	radiometric calibration (essential)
High-res RGB	Sony a7R / integrated 20–61 MP	scouting, ground-truth
Thermal (optional)	FLIR Vue TZ20	crop water stress
LiDAR (optional, biomass/lodging)	Livox / DJI Zenmuse L2	canopy height, biomass

Harvest-due logic: calibrated reflectance → NDVI/NDRE per zone → crop-specific maturity threshold / trained classifier → per-zone "due / not-due / N days" map. Onboard gives an instant flag; the edge gateway produces the full agronomic map.

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7. Ground segment & networking

Item	Example	$
Ground control station	rugged laptop/tablet + **QGroundControl / Mission Planner**	1,500–4,000
Edge gateway box (Tier 2)	Jetson AGX Orin dev/rugged, or x86+GPU	2,000–5,000
RTK base station	Emlid Reach RS3 / RS2+, or NTRIP subscription	2,000–3,500
Field router + PoE switch + UPS	industrial (Teltonika RUTX)	400–900
Power in the field	2 kWh portable power station + solar, or generator	800–2,500

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8. Security & key management (PQC update channel)

Item	Purpose
Secure element on the UAV	Microchip **ATECC608** / TPM 2.0 — stores keys, verifies signed model updates
HSM at the gateway	**YubiHSM 2** or equivalent — signs/verifies **Kyber-1024 / Dilithium** (and GOST R 34.10) model & config updates
Encrypted storage	LUKS on gateway, signed containers for RobustIDPS.ai

This implements the dissertation's **PQC-secured update channel** so models pushed to the drone can't be tampered with in the field.

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9. Tools, materials & test instrumentation

Category	Items
Assembly tools	precision hex/torx drivers, torque driver, soldering station, heat-shrink, Loctite, multimeter, crimpers, zip ties, vibration-damping mounts
RF test gear (validate against the benchmark!)	handheld **spectrum analyzer** (e.g., Signal Hound BB60 / tinySA Ultra) to measure the real J/S and noise floor during the run; SDR (USRP/HackRF) for RF characterisation
Controlled EW emulation	signal generator / GNSS simulator (**LabSat**, Spirent) and jammer emulator — **licensed test range / Faraday enclosure ONLY** (see §11)
Field logistics	rugged transit cases (Pelican), landing pad, tie-downs, tent/shade, ND filters, calibration/reflectance panel, GCP targets + survey markers
Batteries	6–10 smart flight batteries + multi-charger + fireproof LiPo bags

Category	Items
Safety	fire extinguisher (Li-ion rated), first-aid, hi-vis, comms

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10. Software stack — flashing the models onto the chip

1. Flash **JetPack/L4T** on the Jetson; enable max-power then tune the power profile for flight endurance.
2. Convert each model **PyTorch → ONNX → TensorRT INT8**; build the **MambaShield INT8 engine** for the onboard module (calibrate with a field traffic sample).
3. Package **RobustIDPS.ai** as signed Docker containers; onboard runs the detection container, gateway runs the full stack + M1/M6/M7 + federated client.
4. Integrate with the FC over **MAVLink** (MAVSDK/pymavlink); optionally ROS 2 for the sensor/inference graph.
5. Wire the **GNSS C/N_0 + spoof flags** and **radio-link stats** in as live features to M4/M6.
6. Verify the **PQC update path**: sign a test model with Dilithium at the HSM, push, confirm the UAV's secure element accepts only valid signatures.

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11. Regulatory, safety & the EW test caveat

- **Drone ops:** register the aircraft, Remote ID, pilot licence, and a **BVLOS authorization** if flying beyond line of sight; file NOTAMs; carry liability insurance. Requirements differ by country — confirm with your CAA.
- **EW testing is regulated.** Deliberately transmitting GNSS spoofing or jamming **on open spectrum is illegal** in most jurisdictions. Do the jamming/spoofing test run **only**: (a) in a **licensed EW/RF test range**, or (b) inside a **shielded/Faraday enclosure**, or (c) with a **GNSS record & replay simulator** (LabSat/Spirent) injected by cable — never radiated over the air without authorization.
- For the **non-EW agronomy validation**, no special RF licence is needed — fly normally, monitor the link passively, and only *characterise* the ambient J/S with the spectrum analyzer.

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12. Indicative bill of materials (starter test run, one UAV)

#	Subsystem	Representative choice	Qty	Indicative $
1	Airframe + FC	M350 RTK / Freefly Astro + Pixhawk 6X	1	6,000–14,000
2	**Onboard AI chip**	**Jetson Orin NX 16 GB + carrier**	1	700–1,100
3	Onboard co-accel (opt.)	Hailo-8 M.2	1	200–450
4	Anti-spoof GNSS + antenna	Septentrio mosaic-X5 / u-blox ZED-F9P + CRPA	1	500–3,500
5	C2 MANET radio pair	Silvus / Doodle Labs	1 set	3,000–9,000
6	4G/5G + RFD900x backup	modem + telemetry radio	1	400–900
7	Multispectral payload	MicaSense Altum-PT + DLS 2 + panel	1	8,000–13,000
8	RGB / thermal / LiDAR (opt.)	Zenmuse / FLIR / Livox	—	1,500–12,000
9	Edge gateway (Tier 2)	Jetson AGX Orin 64 GB (rugged)	1	2,000–5,000
10	RTK base / NTRIP	Emlid Reach RS3	1	2,000–3,500
11	GCS + networking + power	laptop, router, power station	1 set	3,000–7,000
12	Security (secure element + HSM)	ATECC608 + YubiHSM 2	1 set	700–1,200
13	RF test gear	spectrum analyzer + GNSS sim (rental ok)	1	1,500–20,000
14	Batteries, tools, cases, safety	—	lot	3,000–6,000
	Cloud (Tier 3)	GPU VM for FedGTD + PQC signing	—	~ /month

#	Subsystem	Representative choice	Qty	Indicative $
	Indicative total (1-UAV pilot)			**~ $40k–90k**

Cloud aggregation and the GNSS simulator can be **rented**, and the co-accel, thermal and LiDAR are optional — a lean first test run lands nearer the low end.

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13. Field test-run procedure (one sortie)

1. **Calibrate** — image the reflectance panel; set RTK base / NTRIP; verify secure boot + valid model signatures on the UAV.
2. **Baseline flight** — fly the survey grid; onboard NDVI screen + edge maturity model produce the first **harvest-due map**; RobustIDPS.ai logs a clean link baseline.
3. **Characterise the RF environment** — spectrum analyzer records ambient noise / any interference; log receiver C/N_0 → this is your measured J/S.
4. **(Authorized range only) EW stress** — inject spoofing/jamming via simulator or in the shielded range across a J/S sweep; confirm the UAV holds mission completion in line with the benchmark curve, and that M4/M6/M7 flag and ride through the attack.
5. **Decision + audit** — export the per-zone due/not-due map; export the security event log and the completion-vs-J/S record; compare field results against `UAV-EW-Bench-2026` to validate the models in the real environment.

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14. What each research contribution buys you in the field

Research result	Field benefit
M4 MambaShield INT8 onboard	detects C2/telemetry intrusion in real time, on a low-power chip
M1 CT/SDE-TGNN	catches slow, evolving anomalies across the flight/fleet
M6 uncertainty calibration	avoids false "abort" decisions on ambiguous signals
M7 game-theoretic certificate	a *provable* safe-operating envelope for the operator/regulator
FedGTD federated learning	farms improve the model together without sharing raw data
+17 dB J/S mission-completion margin	the survey **finishes** — and the harvest-due map arrives — even under heavy interference